FORMAL SMOOTHNESS IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4937-4941 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 4943-4963, the complete author proof (19 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Formally smooth morphisms and their lifting criterion, and fpqc descent; the reduction is retained.

Source reference. Original lemma 06CM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4963–4963. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 061K: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4837–4856. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03FX: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 4963–4963. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *The property $\mathcal{P}(f) =$ "f is formally smooth" is fpqc local on the base.*

Proof. Let $f : X \to Y$ be a morphism of algebraic spaces over a scheme S. Choose an index set I and diagrams

$$\begin{array}{ccc} U_i & \xrightarrow[\psi_i]{} & V_i \\ \downarrow & & \downarrow \\ X & \xrightarrow{f} & Y \end{array}$$

with étale vertical arrows and U_i, V_i affine schemes. Moreover, assume that $\coprod U_i \to X$ and $\coprod V_i \to Y$ are surjective, see Properties of Spaces, Lemma 03FX. By Lemma 061K we see that f is formally smooth if and only if each of the morphisms ψ_i are formally smooth. Hence we reduce to the case of a morphism of affine schemes. In this case the result follows from Algebra, Lemma 06CM. Some details omitted. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.